

PRACTICAL WORK 5

Data Classification Models

Dataset (see attached files)

- Breast Cancer Wisconsin (Diagnostic)
 - <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>
- Heart Disease Dataset
 - <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

Learning Objectives

- Understand the workflow of building a classification model in machine learning
- Prepare and preprocess datasets for classification tasks
- Implement classification models using Python and Scikit-learn
- Train models using training datasets
- Evaluate model performance using classification metrics
- Interpret model performance using confusion matrix, precision, recall, and F1 score
- Compare performance of multiple classification algorithms

Task

Complete the following task:

- Data Exploration (visualize feature relationships: histograms, pairplots, correlation heatmaps, etc.)
- Data Preprocessing
 - handle missing values, imputation, duplicate, etc.
 - Feature Scaling/ Normalization (min-max, Z-score)
- Dataset Splitting
 - Training set (70%), validation (10%), Test set (20%)
 - Kfold stratified method (K= 5)
- Implement Classification Models
 - Train at least two classification models.
- Model Prediction (using test dataset)
- Model Evaluation (confusion matrix, accuracy, recall, precision, etc.)
- Model comparison
 - Create a table to compare the performance of the implemented classifiers
 - Discuss which model performs best and why.

Deliverables

- Write a summarized report describing each task results.
 - ❖ Include the screenshots of plots
- Submit the source code (Python Notebook or .py file)